2014 IEEE International Conference on High Performance Computing and Communications (HPCC), 2014 IEEE 6th International Symposium on Cyberspace Safety and Security (CSS) and 2014 IEEE 11th International Conference on Embedded Software and Systems (ICESS)

A Novel Fault Diagnosis in Reversible Logic Circuit

Bikromadittya Mondal
Department of Computer Science and Engineering
B P Poddar Institute of Management and Technology
Kolkata-700052, India
Email: bikmondal@gmail

Susanta Chakraborty
Department of Computer Science and Technology
Indian Institute of Engineering Science and Technology
Shibpur, Howrah-711103, India
Email: susanta.chak@gmail.com, sc@cs.becs.ac.in

***Abstract*—Reversible circuits have widespread applications in quantum computing, low power circuit design, nanotechnology, cryptography etc. The paper addresses the fault diagnosis problem in reversible logic circuit for missing gate fault model. The key idea of this paper is to extract the fault signature for each of the missing gate faults in a reversible circuit. Fault signatures are generated using the successive tree form of Multi-Input Signature Analyzer (MISA). Fault diagnosis is performed by verifying the signature of the circuit under test. The proposed method rapidly identifies the fault locations and the experimental results for benchmark circuits exhibit the efficiency of the technique**.

Keywords—Reversible circuit; Missing gate fault; Fault diagnosis; MISA; Fault signature.

I. INTRODUCTION

Reversible circuit is one of the leading research areas now a day. It has been focused as a special class of quantum circuit which is much faster than non-quantum circuits. Quantum circuits can solve problems like prime factorization, exponentially faster than its classical counterpart. In quantum computing the information is stored in the form of qubit (q-bit) instead of bits that are used in classical circuits (non-quantum). Qubits are microscopic entities such as a photon or atomic spin. A qubit can be in a zero or a one state, conventionally denoted by |0> and |1> respectively. However, it can also be in a superposition of these states, i.e., $\alpha_0|0> + \alpha_1|1>$, where α_0 and α_1 are complex numbers called amplitude [1, 2, 3, 5].

Automated synthesis and testing of reversible circuits have attracted as a major research challenge. Fault models and testing approaches [7-10] are proposed by researchers in recent years. Automatic Test Pattern Generation is an electronic design automation method used to find an input (or test) sequence that, when applied to a digital circuit, enables automatic test equipment to distinguish between the correct circuit behavior and the faulty circuit behavior caused by defects. ATPG based techniques for reversible circuits are introduced by R. Wille et al. [14]. But to the best of our knowledge very few works are done in the area of fault diagnosis in reversible circuit so far. The aim of fault diagnosis is to find out the location of faults. To perform an efficient diagnosis, a test set with good diagnostic capability is needed. Therefore, diagnostic ATPG is a critical part of fault diagnosis. In diagnostic test generation, the goal is to find a test sequence such that the circuit produces a different response for different faults present in the circuit. Therefore the problem of diagnostic test generation is much harder than conventional test generation. Fault localization in reversible circuit by using adaptive tree is introduced in [8]. Another approach generates fault localization tree to locate the faults. Both the techniques consider all possible input assignments to identify the faulty behavior of the circuit. Error debugging has been introduced in [12, 13]. Fault masking using majority voters with error masking capabilities for reversible circuits is presented in [17]. A new fault diagnosis technique [18] is proposed that explicitly exploits the advantageous properties of reversible circuit.

In this paper, the problem of the fault diagnosis in reversible logic circuit is addressed under missing gate fault model. The proposed fault diagnosis technique is used to locate the presence of faults using successive tree structure of Multi Input Signature Analyzer. Pseudorandom test vectors are applied to the circuit and from the output responses of the faulty and the fault free circuit a one dimensional fault signature is generated. Fault diagnosis is performed by verifying the signature of the circuit under test. The proposed method rapidly identifies the location of any missing gate fault present in the circuit.

The structure of the rest of the paper is as follows: a brief idea of reversible logic circuit and reversible logic gates and their basic properties are discussed and the model of missing control faults is introduced in section II. The proposed fault diagnosis technique with illustration is presented in section III. Experimental results are shown and conclusions are drawn in section IV and section V respectively.

II. PRELIMINARIES

In this section the basic properties of reversible circuit and reversible logic gates are presented. The behaviors of the different missing gate fault models are also discussed.

A. Reversible Logic Gate/Circuit

A circuit is said to be reversible if the (Boolean) function it computes is bijective i.e., there is one-to-one and onto correspondence between its inputs and outputs. The input vector can be uniquely determined from the output vector and vice versa. A necessary condition is that the circuit has the same number of input and output wires. No fanout is allowed and it is a loop free circuit. An $n \times n$ reversible logic circuit $G = g_1 g_2 g_3 ... g_d$ is a cascade of d reversible logic gates. Most of the classical gate like AND, OR, XOR are not reversible. Controlled–NOT (CNOT) gate proposed by Feynman, Toffoli gate, and Fredkin gate are well known reversible logic gate. Fig.1(a) shows a 3×3 reversible circuit containing 6 reversible logic gates. It

978-1-4799-6123-8/14 $31.00 © 2014 IEEE
DOI 10.1109/HPCC.2014.125

computes the bijective function (0, 1, 2, 3, 4, 5, 6, 7) → (7, 0, 1, 3, 4, 2, 6, 5).

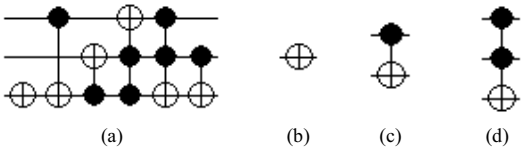

Fig.1(a) Reversible Circuit (b) NOT gate (c) CNOT gate (d) 2-CNOT gate

The *Reversible NOT* gate (Fig. 1(b)) is a 1-input 1-output (1×1) gate. The bijective function is (x) → NOT (x). The *Controlled-NOT* (also called *CNOT*) gate (Fig. 1(c)) is a 2×2 gate. The bijective function is (x, y) → (x, x⊕y). The *Toffoli* (also called *2-CNOT*) gate (Fig. 1(d)) is a 3×3 gate. The bijective function is (x, y, z) → (x, y, xy⊕z). The *N-CNOT* gate is a logical extension of 2-CNOT gate. It computes the bijective function (x_1, x_2, …x_n) → (x_1, x_2, … x_{n-1}, x_1x_2 … x_{n-1}⊕x_n).

B. Missing Gate fault

A *single missing gate fault* (*SMGF*) in a reversible circuit is defined as the complete disappearance of a reversible gate from the circuit. The *multiple missing gate fault (MMGF)* is defined as the disappearance of two or more CNOT gates. *Partially missing gate fault (PMGF)* model is defined as the partially misaligned or mistuned of any CNOT gate.

III. PROPOSED WORK

A. Test Structure

The test structure is shown in Fig.2. The Test Input Generator (TIG) can generate pseudorandom test inputs and also can be used as an external input generator. The second block is the reversible circuit under test (CUT). The circuit has m inputs and m outputs. Multi input signature analyzer (MISA) receives m inputs from the CUT and provides single output that is monitored during the diagnosis process.

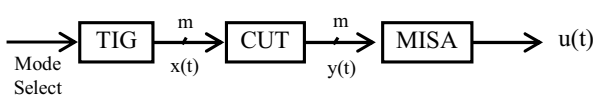

Fig.2 Test structure of fault diagnosis

B. Design of MISA

In this section we propose the structure of the Multi-Input Signature Analyzer that is used for fault diagnosis. Fig.3 shows the design of an n-input 1-output MISA.

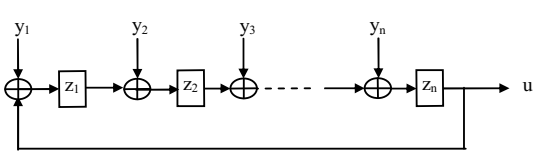

Fig.3 Design of an n-input 1-output MISA

The n parallel inputs are y_1, y_2, ….,y_n and the single output is u. The n number of temporary storage units z_1, z_2, …,z_n are used to hold the intermediate signatures at any instance. The outputs of the storage units are described as follows:

$$z_1(t+1) = y_1(t) \oplus z_n(t),$$
$$z_i(t+1) = y_i(t) \oplus z_{i-1}(t) \text{ and}$$
$$u(t) = z_n(t)$$

The state transitions of a 3-input 1-output MISA are shown in Table I. PS indicates the present states at instance t, IN indicates the external inputs and NS indicates the next states at instance t+1.

TABLE I: STATE TRANSITION

	NS		
PS \ IN	$y_1(t)$	$y_2(t)$	$y_3(t)$
$z_1(t)$	-	$z_2(t+1)$	-
$z_2(t)$	-	-	$z_3(t+1)$
$z_3(t)$	$z_1(t+1)$	-	-

The operations of MISA can be described in the form of a successor tree. The successor tree of a 3 input 1 output MISA is shown in Fig.4. $z_1(t)$, $z_2(t)$ and $z_3(t)$ are the states of the three temporary signature holders at time t. $z_i(t+1)$, $z_i(t+2)$, ….are the states of the ith storage holders at instance (t+1), (t+2), …..etc. These states are activated by the states in the previous stage and by the influence of the external inputs supplied to the circuit.

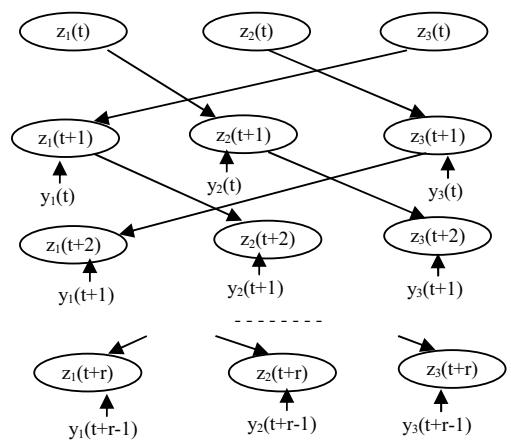

Fig.4 Successive tree

The one dimensional output u(t) at time t is as follows:

$$\begin{aligned} u(t) &= z_n(t) \\ &= z_{n-1}(t-1) \oplus y_n(t-1) \\ &= z_{n-2}(t-2) \oplus y_{n-1}(t-2) \oplus y_n(t-1) \\ &= z_{n-3}(t-3) \oplus y_{n-2}(t-3) \oplus y_{n-1}(t-2) \oplus y_n(t-1) \\ &\ldots\ldots\ldots\ldots \\ &= z_1(t-n) \oplus y_1(t-n) \oplus y_2(t-n-1) \oplus .. \oplus y_n(t-1) \end{aligned}$$

C. Proposed Algorithm

The proposed diagnosis algorithm is as follows:

1. The TIG generates a sequence of T test inputs x(0), x(1),……, x(T-1) and these test inputs are fed to the reversible circuit. The outputs of the reversible circuit y(0), y(1), ……, y(T-1) are fed to the inputs of the MISA that is expressed in the successive tree form as in Fig.8. The one dimensional output u(0), u(1), ….., u(T-1) are preserved.
2. Now we assume a missing gate fault in the reversible circuit. The TIG generates the same test inputs x(0), x(1), ….. as in step1 and the corresponding test responses are again

accumulated by the MISA. This time the one dimensional outputs are continuously monitored until we get a different output than the fault free circuit i.e., $u_c(k) \neq u_f(k)$, where $u_c(k)$ is the output of the MISA for fault free circuit at time k and $u_f(k)$ is the output of the MISA for faulty circuit at time k.

3. From step2, the last test vector x(k) is repeatedly applied (maximum n times) to both the faulty and fault free circuit and the outputs of the MISA are compared until we again get a different output. If the same input vector $(y_1, y_2,\ldots,y_n)$ is applied n times to the MISA in the state $(z_1, z_2,\ldots,z_n)$ then sequence of outputs of length n i.e., $(u(t+1),\ldots,u(t+n))$ be shown as

$$u(t+1) = y_n \oplus z_{n-1}$$
$$u(t+2) = y_n \oplus y_{n-1} \oplus z_{n-2}$$
$$\ldots\ldots\ldots$$
$$u(t+n) = y_n \oplus y_{n-1} \oplus \ldots\ldots \oplus y_1 \oplus z_n$$

The one dimensional outputs are stored as a signature of the corresponding missing gate fault. Thus the signatures for each of the missing gate faults are extracted and stored for fault diagnosis.

4. In the fault diagnosis phase, the inputs x(0), x(1), ….. as in step1 are applied to the CUT and the accumulated outputs of the MISA are compared with the outputs of the fault free circuit. If the outputs are same then the CUT is considered to be fault free and no further diagnosis is required. Otherwise, the last test vector is repeatedly applied and the one dimensional signature is verified with the signature of the stored faults to identify the location of the fault.

D. Example of Fault Diagnosis

The proposed technique is applied on the reversible circuit shown in Fig.5. The 3×3 circuit contains three reversible gates. The three outputs of the TIG are connected to the three inputs of the CUT. The outputs of the CUT are connected to the three inputs of the MISA.

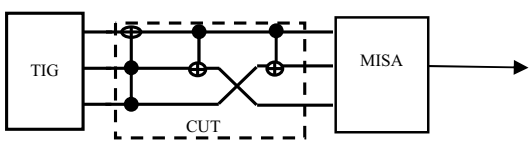

Fig.5 Reversible Circuit Under Test

Pseudorandom test patterns 010, 010, 110, 010, ... are generated by the TIG and fed to the fault free circuit. The output responses 001, 001, 110, 001, ….of the circuit are fed to the three inputs of the MISA. In Table II, the first column t shows the instance of the test vector. The second column $x_3x_2x_1$ and third column $y_3y_2y_1$ show the input and output responses respectively. The last column shows the output u_c of the MISA. The one dimensional output of the MISA, 0000001….. is preserved for the fault diagnosis.

TABLE II: INPUT AND OUTPUT RESPONSES

t	$x_3\,x_2\,x_1$	$y_3\,y_2\,y_1$	u_c
0	010	001	0
1	010	001	0
2	110	110	0
3	010	001	0
4	000	000	0
5	101	101	0
6	011	100	1
.	…	…	.

Now we assume the first CNOT gate is faulty i.e., missing. As a result, the CUT contains only two CNOT gates. The faulty circuit is shown in Fig.6.

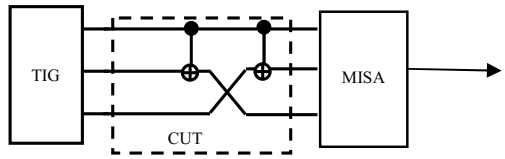

Fig.6 Single Missing gate fault

Next the same set of test inputs is applied to the faulty CUT. The output response of the CUT will be different for some inputs in the presence of the fault. The input and output responses of the fault free and faulty CUT are shown in Table III. The first three columns of Table III are similar to the Table II i.e., the time instances, input vectors and output vectors respectively. The fourth column $y_{f3}y_{f2}y_{f1}$ shows the output responses of the faulty CUT. The last two columns u_c and u_f show the one dimensional output of the MISA for fault free and faulty circuit respectively. This time MISA outputs are continuously monitored and u_c is compared with u_f. The pseudorandom inputs are applied until we get a first time different output at time t. In Table III we find that at t = 6, u_c and u_f are different.

TABLE III: FAULT SIGNATURE

t	$x_3\,x_2\,x_1$	$y_3\,y_2\,y_1$	$y_{f3}\,y_{f2}\,y_{f1}$	u_c	u_f
0	010	001	001	0	0
1	010	001	001	0	0
2	110	110	110	0	0
3	010	001	001	0	0
4	000	000	000	0	0
5	101	101	101	0	0
6	011	100	011	**1**	**0**
7	011	100	011	0	0
8	011	100	011	**1**	**0**

Finally, at this point the last test input vector 011 i.e., the input at t = 6, is repeatedly applied to both the fault free and faulty CUT. These times also the one dimensional output sequence of the MISA for fault free and faulty circuit is continuously compared. The repetition is continued until we again get a different output. In the Table III at t = 8, different outputs are observed. Here only two repetitions of the last test vector 011 are required to confirm the fault. The fault signature 101 and 000 are recorded for use during the fault diagnosis. In case of no confirmation of the fault, the repetition is continued to maximum n times, where n is the number of input lines of the circuit.

The diagnosis process is very simple. The same test patterns are applied to the reversible CUT. If the one dimensional outputs of the MISA are similar to the MISA outputs of the fault free CUT then the circuit is considered to be fault free and no further diagnosis is required. But if any deviation of the MISA output is observed at time t, then it indicates that a fault has occurred. To confirm the fault from time t+1, the last test input is repeatedly applied

and the one dimensional signature is compared with the previously recorded signature. Therefore, to ensure the diagnosis of every fault, fault signature for each fault is prepared.

IV. EXPERIMENTAL RESULTS

Experimental results of our proposed method are shown in Table IV. The first column shows the benchmark reversible circuits that are available at http://webhome.cs.uvic.ca/~dmaslov/. Next two columns N and n are the number of reversible gates and number of inputs respectively in the circuit. LEN is the maximum length of the test sequence required to detect all the faults of a particular missing gate fault model. REP is the maximum number of repetitions of the last test vector required to confirm all the fault locations. The proposed method has been applied for single missing gate faults as well as multiple and partial missing gate faults.

TABLE IV: EXPERIMENTAL RESULTS

CIRCUIT	N	n	SINGLE		MULTIPLE		PARTIAL	
			LEN	REP	LEN	REP	LEN	REP
3_17tc	6	3	5	3	8	3	6	3
Ham3tc	5	3	8	3	7	3	8	3
Ham7tc	23	7	22	5	15	7	22	7
Ham7-25-49	24	7	14	5	12	7	25	5
6symd2	20	10	37	10	14	10	34	10
Rd73d2	20	10	15	10	22	9	18	10
9symd2	28	12	48	12	67	12	81	12
Ham15tc1	132	15	97	13	79	12	112	13
Rd84d1	28	15	166	15	153	15	164	15

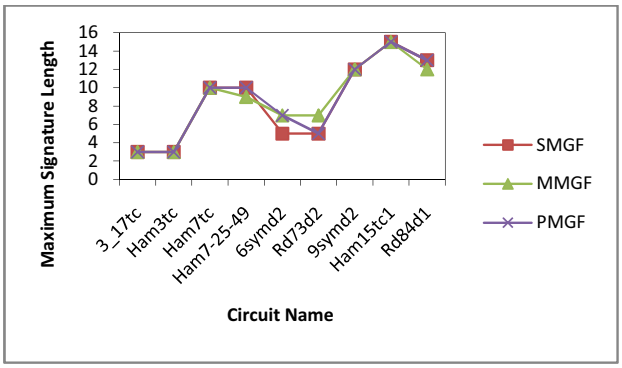

Fig.7 Comparison of maximum signature lengths for benchmark circuits

Fig.7 shows the comparison of maximum signature length for the diagnosis of single, multiple and partial missing gate faults for benchmark reversible circuits.

V. CONCLUSIONS

This paper proposes a novel technique of fault diagnosis in a reversible logic circuit using Multi-Input Signature Analyzer. Fault signatures for missing gate faults are generated from the successive tree structure of MISA. Fault diagnosis is done by verifying the signature of the faulty circuit with the recorded signatures. The technique is simple and fast. Experimental results of benchmarks reversible circuits show that, the proposed technique is very effective for fault diagnosis of reversible circuits for single, multiple and partial missing gate faults. The method may be extended in future for fault diagnosis in other fault models in reversible circuits.

REFERENCES

[1] R. Landauer, "Irreversibility and heat generation in the computing process," *IBM Journal of Research and Development,* vol. 5, 1961, p.183.

[2] T. Toffoli, "Reversible computing," in *Automata, Languages and Programming*, W. de Bakker and J. van Leeuwen, Eds. Springer, 1980, p. 632, technical Memo MIT/LCS/TM-151, MIT Lab. for Computer Science.

[3] A. Peres, "Reversible logic and quantum computers," *Phys. Rev. A*, no. 32, 1985, pp. 3266–3276.

[4] P. W. Shor, "Algorithms for quantum computation: discrete logarithms and factoring," *Foundations of Computer Science*, 1994, pp. 124–134.

[5] M. Nielsen and I. Chuang, *Quantum Computation and Quantum Information*. Cambridge Univ. Press, 2000.

[6] L. M. K. Vandersypen, M. Steffen, G. Breyta, C. S. Yannoni, M. H. Sherwood, and I. L. Chuang, "Experimental realization of Shor's quantum factoring algorithm using nuclear magnetic resonance," *Nature*, vol. 414, 2001, p. 883.

[7] J. P. Hayes, I. Polian, and B. Becker, "Testing for missing-gate faults in reversible circuits," in *Asian Test Symp.*, 2004, pp. 100–105.

[8] K. Ramasamy, R. Tagare, E. Perkins, and M. Perkowski, "Fault localization in reversible circuits is easier than for classical circuits," in *Workshop on Logic and Synthesis*, 2004.

[9] I. Polian, T. Fiehn, B. Becker, and J. P. Hayes, "A family of logical fault models for reversible circuits," in *Asian Test Symp.*, 2005, pp. 422–427.

[10] D. Pierce, J. Biamonte, and M. Perkowski, "Test set generation and fault localization software for reversible circuits," in *Int'l Symposium on Representations and Methodologies for Emergent Computing Technologies*, September 2005.

[11] R. Wille, D. Große, L. Teuber, G. W. Dueck, and R. Drechsler, "RevLib: an online resource for reversible functions and reversible circuits," in *Int'l Symp. on Multi-Valued Logic*, 2008, pp. 220–225, RevLib is available at http://www.revlib.org.

[12] R. Wille, D. Große, S. Frehse, G. W. Dueck, and R. Drechsler, "Debugging of Toffoli networks," in *Design, Automation and Test in Europe*, 2009, pp. 1284–1289.

[13] J. C. Jung, S. Frehse, R. Wille, and R. Drechsler, "Enhancing debugging of multiple missing control errors in reversible logic," in *Great Lakes Symp. VLSI*, 2010, pp. 465–470.

[14] R. Wille, H. Zhang, and R. Drechsler, "ATPG for reversible circuits using simulation, Boolean satisfiability, and pseudo Boolean optimization," in *IEEE Annual Symposium on VLSI*, 2011, pp. 120–125.

[15] A. Paler, A. Alaghi, I. Polian, and J. P. Hayes, "Tomographic testing and validation of probabilistic circuits," in *European Test Symp.*, 2011, pp. 63–68.

[16] H. Rahaman, D. K. Kole, D. K. Das, B. B. Bhattacharya, "Fault diagnosis in reversible circuits under missing-gate fault model," *Computers & Electrical Engineering*, Volume 37, Issue 4, July 2011, pp. 475-485.

[17] M. Zamani, N. Farazmand, M. B. Tahoori, "Fault Masking and Diagnosis in Reversible Circuits," *16th IEEE European Test Symposium*, 2011, pp.69-74.

[18] Z. Hongyan, R. Wille, R. Drechsler, "Improved Fault Diagnosis for Reversible Circuits," *20th Asian Test Symposium*, 2011, pp. 207-212.

[19] M. Lukac, M. Kameyama, M. Perkowski, P. Kerntopf, "Analysis of Reversible and Quantum Finite State Machines Using Homing, Synchronizing and Distinguishing Input Sequences", in IEEE 43rd International Symposium on Multiple-Valued Logic (ISMVL), 2013, pp. 322-327.